

Equations Implemented in RIPTA

A0 Preliminaries

Table A1 summarizes the notation used throughout this document. All statistics are estimated from data; hatted symbols denote estimators. Population labels P_1, P_2, P_3 and outgroup O are reserved for the Introgression section (**Section A2**), where they carry fixed model-based roles. Generic population labels P_A, P_B, P_C, P_D, P_E are used in the f -statistics section (**Section A3**). The ancestral (A) and derived (B) allele notation appears only in the Introgression section; all other sections use reference and alternative alleles.

Table A1: **Summary of the notation used throughout this document.**

Each symbol used throughout this document is listed with a brief description.

Symbol	Description
L	Total sequence length (number of sites).
M	Number of biallelic segregating sites in the genome (or in a genomic window), indexed by $m = 1, \dots, M$.
P_x	Generic population label for population x .
A / B	Ancestral and derived allele, polarized by the outgroup O ; used only in the Introgression section.
$P_x(m)$	Allele observed in population P_x at site m when one sequence is sampled; $P_x(m) \in \{A, B\}$.
$O(m)$	Allele observed in the outgroup O at site m .
$c_{m,x}$	Total number of chromosomes observed at biallelic site m in population P_x . Equals $2 \times$ the number of diploid individuals (or $1 \times$ for pseudohaploid individuals) with called genotypes; varies across sites in the presence of missing data.

Continued on next page

Table A1: **Summary of the notation used throughout this document.**

Each symbol used throughout this document is listed with a brief description. (Continued)

Symbol	Description
$n_{m,x}^{(\text{aac})}$	Alternative allele count at site m in population P_x . The reference allele count is $c_{m,x} - n_{m,x}^{(\text{aac})}$.
$n_{m,x}^{(\text{dac})}$	Derived allele count at site m in population P_x . The ancestral allele count is $c_{m,x} - n_{m,x}^{(\text{dac})}$.
$\hat{p}_{m,x}^{(\text{aaf})}$	Alternative allele frequency: $n_{m,x}^{(\text{aac})}/c_{m,x}$. The reference allele frequency is $1 - \hat{p}_{m,x}^{(\text{aaf})}$.
$\hat{p}_{m,x}^{(\text{daf})}$	Derived allele frequency: $n_{m,x}^{(\text{dac})}/c_{m,x}$. The ancestral allele frequency is $1 - \hat{p}_{m,x}^{(\text{daf})}$.
$h_{m,x}$	Sample half-heterozygosity at site m in population P_x : $\hat{p}_{m,x}^{(\text{aaf})}(1 - \hat{p}_{m,x}^{(\text{aaf})})$. Assumes large sample size and no missing data.
$\hat{h}_{m,x}$	Unbiased half-heterozygosity at site m in population P_x : $n_{m,x}^{(\text{aac})}(c_{m,x} - n_{m,x}^{(\text{aac})}) / c_{m,x}(c_{m,x} - 1)$.
h_x	Average sample half-heterozygosity over M segregating sites: $\frac{1}{M} \sum_{m=1}^M h_{m,x}$.
$\hat{h}_x$	Average unbiased half-heterozygosity over M segregating sites: $\frac{1}{M} \sum_{m=1}^M \hat{h}_{m,x}$.
$\pi_{m,x}$	Sample nucleotide diversity at site m in P_x : $2h_{m,x}$. Assumes large sample size and no missing data.
π_x	Genome-wide (or window-based) sample nucleotide diversity: $\frac{1}{M} \sum_{m=1}^M \pi_{m,x}$. Assumes large sample size and no missing data.
$\delta_{m,x}, \kappa_{m,x}$	Within-population pairwise differences and comparisons at site m used in the pixy $\hat{\pi}$ estimator.
$\hat{\pi}_x$	Genome-wide (or window-based) unbiased nucleotide diversity: ratio of total differences to total comparisons [1].
d_{m,P_x-P_y}	Absolute divergence at site m between P_x and P_y . Assumes large sample size and no missing data.
$d_{P_x-P_y}$	Genome-wide (or window-based) absolute divergence. Assumes large sample size and no missing data.
$\delta_{m,xy}, \kappa_{m,xy}$	Between-population pairwise differences and comparisons at site m used in the pixy $\hat{d}_{P_x-P_y}$ estimator.

Continued on next page

Table A1: **Summary of the notation used throughout this document.**

Each symbol used throughout this document is listed with a brief description. (Continued)

Symbol	Description
$\hat{d}_{P_x-P_y}$	Genome-wide (or window-based) unbiased absolute divergence: ratio of total differences to total comparisons [1].
$F_{ST}(P_x, P_y)$	Large sample size estimator of Hudson's F_{ST} between populations P_x and P_y .
$\hat{F}_{ST}(P_x, P_y)$	Unbiased Hudson's F_{ST} estimator between populations P_x and P_y [2]. Corrected for finite sample size differences.
$\mathbf{1}_{\text{site pattern}}(m)$	Indicator variable to denote that a given site pattern is observed at site m .
$\hat{n}_{\text{site pattern}}$	Estimated site pattern count from sequence data for site pattern $\in \{BBAA, BABA, ABBA, BAAA, ABAA, AABA\}$.
$\mathbf{1}_{P_x \neq P_y}(m)$	Indicator variable to denote that an allelic difference is observed between the sampled sequences from P_x and P_y at site m .
$\hat{D}(P_1, P_2, P_3, O)$	Test statistic for studying introgression using the <i>ABBA</i> and <i>BABA</i> site patterns.
$\hat{D}_{anc}(P_1, P_2, P_3, O)$	Test statistic for studying introgression using the <i>BAAA</i> and <i>ABAA</i> site patterns.
$\hat{D}^+(P_1, P_2, P_3, O)$	Test statistic for studying introgression using the <i>ABBA</i> , <i>BABA</i> , <i>BAAA</i> , and <i>ABAA</i> site patterns.
$\hat{f}_{hom}(P_1, P_2, P_3, O)$	Test statistic for studying introgression using the <i>ABBA</i> and <i>BABA</i> site patterns.
$\hat{S}(P_1, P_x, P_3, O)$	Difference in derived allele sharing between P_1 and P_x compared to P_1 and P_3 .
$\hat{D}_p(P_1, P_2, P_3, O)$	Test statistic for studying introgression using all derived allele sharing site patterns.
$\hat{d}_f(P_1, P_2, P_3, O)$	Test statistic for studying introgression using all derived allele sharing site patterns.
$\hat{D}_3(P_1, P_2, P_3)$	Test statistic for studying introgression using measures of absolute divergence.
$\hat{D}^*(P_1, P_2, P_3, O)$	Test statistic for studying introgression using all derived allele sharing site patterns.
$f_2(P_A, P_B)$	f_2 estimator quantifying genetic drift separating P_A and P_B . Assumes large sample size and no missing data.

Continued on next page

Table A1: **Summary of the notation used throughout this document.**

Each symbol used throughout this document is listed with a brief description. (Continued)

Symbol	Description
$\hat{f}_2(P_A, P_B)$	Unbiased f_2 estimator. Corrected for finite sample size.
$f_3(P_C; P_A, P_B)$	f_3 estimator quantifying genetic drift on an unrooted three-population tree separating the target population P_C from the two reference populations P_A and P_B . Assumes large sample size and no missing data.
$\hat{f}_3(P_C; P_A, P_B)$	Unbiased f_3 estimator; significantly negative values suggest P_C is admixed from P_A and P_B . Corrected for finite sample size.
$\hat{f}_3^*(P_C; P_A, P_B)$	Normalized f_3 : $\hat{f}_3(P_C; P_A, P_B) / 2\hat{h}_{P_C}$.
$\hat{f}_4(P_A, P_B; P_C, P_D)$	f_4 estimator; a test of treeness for the unrooted four-population tree $((P_A, P_B), (P_C, P_D))$. Assumes large sample size and no missing data.
$\hat{\alpha}$	f_4 -ratio estimate of the admixture proportion. No bias correction needed.

A0.1 Sequence Length and Biallelic Sites

Consider a genomic region of total length L sites. Among these, let M denote the number of *biallelic segregating sites* in the genome (or in a genomic window), *i.e.*, sites at which exactly two alleles are observed across all populations under consideration. All summations indexed by $m = 1, \dots, M$ range over these biallelic sites only.

A0.2 Reference and Alternative vs. Ancestral and Derived Alleles

At each biallelic site m , the two observed alleles are designated *reference* and *alternative* according to the encoding in the input data (*e.g.*, a VCF file). In the Introgression section (Section A2), alleles are further polarized using the outgroup O , where the allele carried by O is denoted as the *ancestral* allele (A) and the other allele is denoted as the *derived* allele (B); note that D_3 is the one exception that does not require an outgroup and therefore uses reference/alternative alleles. The Diversity and Differentiation (Section A1) and f -statistics (Section A3) sections use reference/alternative alleles exclusively and do not require an outgroup.

A0.3 Allele Counts and Frequencies

Let $c_{m,x}$ denote the total number of chromosomes observed at biallelic site m in population P_x . For diploid individuals with called genotypes, $c_{m,x}$ equals twice the number of individuals; for pseudohaploid individuals, $c_{m,x}$ equals the number of individuals. In the pres-

ence of missing data, both allele counts and $c_{m,x}$ depend on the number of called genotypes at that site and may therefore vary across sites.

Let $n_{m,x}^{(\text{aac})}$ denote the number of alternative alleles observed at site m in population P_x . The number of reference alleles is $c_{m,x} - n_{m,x}^{(\text{aac})}$. The estimated alternative allele frequency is

$$\hat{p}_{m,x}^{(\text{aaf})} = \frac{n_{m,x}^{(\text{aac})}}{c_{m,x}}, \quad (\text{A0.1})$$

and the reference allele frequency is $1 - \hat{p}_{m,x}^{(\text{aaf})}$.

When the outgroup O is used to polarize alleles, let $n_{m,x}^{(\text{dac})}$ denote the number of derived alleles at site m in population P_x . The number of ancestral alleles is $c_{m,x} - n_{m,x}^{(\text{dac})}$. The estimated derived allele frequency is

$$\hat{p}_{m,x}^{(\text{daf})} = \frac{n_{m,x}^{(\text{dac})}}{c_{m,x}}, \quad (\text{A0.2})$$

and the ancestral allele frequency is $1 - \hat{p}_{m,x}^{(\text{daf})}$.

A0.4 Allele Counts from Genotype Probabilities

When genotype probabilities (GP) are available instead of hard-called genotypes, allele counts are computed as expected counts from the posterior genotype distribution. Let $\text{Pr}(\text{Homozygous Reference})$, $\text{Pr}(\text{Heterozygous})$, and $\text{Pr}(\text{Homozygous Alternative})$ denote the posterior probabilities of the three diploid genotype classes at site m for a single individual in population P_x . The expected reference and alternative allele counts contributed by that individual are

$$\text{rac} = 2 \text{Pr}(\text{Homozygous Reference}) + \text{Pr}(\text{Heterozygous}), \quad (\text{A0.3a})$$

$$\text{aac} = 2 \text{Pr}(\text{Homozygous Alternative}) + \text{Pr}(\text{Heterozygous}), \quad (\text{A0.3b})$$

and these expected counts are summed across all individuals in the population to obtain $n_{m,x}^{(\text{aac})}$ and $c_{m,x}$.

A0.5 Pseudohaploid Estimation

For pseudohaploid data, a single allele is drawn at random from each individual's genotype at each site, so that each individual contributes one chromosome to $c_{m,x}$. This affects the bias-correction terms in unbiased estimators: the denominator $c_{m,x} - 1$ in the unbiased half-heterozygosity (Eq. (A0.5)) and in the f_2 and f_3 estimators (Section A3) correctly accounts for the reduced number of independently sampled alleles.

A0.6 Half-heterozygosity

The sample half-heterozygosity at site m in population P_x is

$$h_{m,x} = \hat{p}_{m,x}^{(\text{aaf})} (1 - \hat{p}_{m,x}^{(\text{aaf})}) . \quad (\text{A0.4})$$

This is a biased estimate of the population quantity $p_{m,x}^{(\text{aaf})} (1 - p_{m,x}^{(\text{aaf})})$. An unbiased estimator is

$$\hat{h}_{m,x} = \frac{n_{m,x}^{(\text{aac})} (c_{m,x} - n_{m,x}^{(\text{aac})})}{c_{m,x} (c_{m,x} - 1)} . \quad (\text{A0.5})$$

Genome-wide (or window-based) averages over M segregating sites are

$$h_x = \frac{1}{M} \sum_{m=1}^M h_{m,x} , \quad (\text{A0.6a})$$

$$\hat{h}_x = \frac{1}{M} \sum_{m=1}^M \hat{h}_{m,x} . \quad (\text{A0.6b})$$

A1 Diversity and Differentiation

All statistics in this section are computed from alternative allele frequencies $\hat{p}_{m,x}^{(\text{aaf})}$ and do not require an outgroup. Note, that in the current implementation of RIPTA, the statistics here are computed over segregating sites only and that they are invariant to which allele is labelled reference or alternative, and therefore does not depend on outgroup polarization.

A1.1 Nucleotide Diversity

At a single biallelic site m , the sample nucleotide diversity within population P_x is

$$\pi_{m,x} = 2h_{m,x} = 2 \left(\hat{p}_{m,x}^{(\text{aaf})} (1 - \hat{p}_{m,x}^{(\text{aaf})}) \right), \quad (\text{A1.1})$$

and the genome-wide (or window-based) nucleotide diversity is

$$\pi_x = \frac{1}{M} \sum_{m=1}^M \pi_{m,x}. \quad (\text{A1.2})$$

Eqs. (A1.1) and (A1.2) assume large sample sizes and no missing data. To handle missing data and finite sample sizes, [1] compute $\hat{\pi}$ by counting the number of pairwise allelic differences and the number of pairwise allelic comparisons at each site, then taking the ratio of sums. At a single biallelic site m in population P_x , the number of pairwise differences and pairwise comparisons are

$$\delta_{m,x} = n_{m,x}^{(\text{aac})} (c_{m,x} - n_{m,x}^{(\text{aac})}), \quad (\text{A1.3a})$$

$$\kappa_{m,x} = \binom{c_{m,x}}{2} = \frac{c_{m,x}(c_{m,x} - 1)}{2}, \quad (\text{A1.3b})$$

where $\delta_{m,x}$ counts the number of within-population allelic comparisons that differ (equivalently, the product of the two allele counts), and $\kappa_{m,x}$ counts the total number of within-population pairwise comparisons among the $c_{m,x}$ observed alleles. The genome-wide (or window-based) unbiased nucleotide diversity is the ratio of total differences to total comparisons across all sites with at least two observed alleles [1]:

$$\hat{\pi}_x = \frac{\sum_{m=1}^M \delta_{m,x}}{\sum_{m=1}^M \kappa_{m,x}}. \quad (\text{A1.4})$$

Sites with fewer than two called alleles contribute to neither the numerator nor the denominator. Note that for a single site, the per-site ratio $\delta_{m,x}/\kappa_{m,x} = 2\hat{h}_{m,x}$, where $\hat{h}_{m,x}$ is the unbiased half-heterozygosity from Eq. (A0.5); however, the genome-wide `pixy` estimator in Eq. (A1.4) is a ratio of sums, not an average of per-site ratios.

A1.2 Absolute Differentiation

For the simplest case of one (haploid) sequence per population, define the indicator variable

$$\mathbf{1}_{P_x \neq P_y}(m) = \begin{cases} 1 & \text{if } P_x(m) \neq P_y(m), \\ 0 & \text{otherwise,} \end{cases} \quad (\text{A1.5})$$

so that $d_{P_x - P_y} = \sum_{m=1}^M \mathbf{1}_{P_x \neq P_y}(m)$. Given multiple sequences from the two focal populations, the large sample size allele frequency analogue at a single site is

$$d_{m, P_x - P_y} = \hat{p}_{m,x}^{(\text{aaf})} (1 - \hat{p}_{m,y}^{(\text{aaf})}) + \hat{p}_{m,y}^{(\text{aaf})} (1 - \hat{p}_{m,x}^{(\text{aaf})}), \quad (\text{A1.6})$$

and the large sample size genome-wide (or window-based) estimate is

$$d_{P_x - P_y} = \frac{1}{M} \sum_{m=1}^M d_{m, P_x - P_y}. \quad (\text{A1.7})$$

Note that Eqs. (A1.6) and (A1.7) assume large sample sizes and no missing data.

To handle missing data and finite sample sizes, [1] compute $\hat{d}_{P_x - P_y}$ by counting the number of pairwise allelic differences *between* populations and the number of pairwise allelic comparisons between populations at each site, then taking the ratio of sums. At a single biallelic site m , the number of between-population differences and comparisons are

$$\delta_{m,xy} = n_{m,x}^{(\text{aac})} (c_{m,y} - n_{m,y}^{(\text{aac})}) + n_{m,y}^{(\text{aac})} (c_{m,x} - n_{m,x}^{(\text{aac})}), \quad (\text{A1.8a})$$

$$\kappa_{m,xy} = c_{m,x} \times c_{m,y}, \quad (\text{A1.8b})$$

where $\delta_{m,xy}$ counts the number of between-population allelic comparisons that differ and $\kappa_{m,xy}$ counts the total number of between-population pairwise comparisons. The genome-wide (or window-based) estimate is the ratio of total differences to total comparisons across all sites where both populations have at least one observed allele [1]:

$$\hat{d}_{P_x-P_y} = \frac{\sum_{m=1}^M \delta_{m,xy}}{\sum_{m=1}^M \kappa_{m,xy}} . \quad (\text{A1.9})$$

Sites where either population has no called alleles contribute to neither the numerator nor the denominator. Note that for a single site the per-site ratio $\delta_{m,xy}/\kappa_{m,xy}$ equals d_{m,P_x-P_y} from Eq. (A1.6); however, the genome-wide pixy estimator in Eq. (A1.9) is a ratio of sums, not an average of per-site ratios.

A1.3 Relative Differentiation

Hudson's estimator of F_{ST} [2, 3] measures population differentiation as the ratio of between-population variance to total variance in allele frequencies. For large sample sizes, the per-site numerator and denominator are

$$N_m = (\hat{p}_{m,x}^{(\text{aaf})} - \hat{p}_{m,y}^{(\text{aaf})})^2 , \quad (\text{A1.10a})$$

$$D_m = \hat{p}_{m,x}^{(\text{aaf})} (1 - \hat{p}_{m,y}^{(\text{aaf})}) + \hat{p}_{m,y}^{(\text{aaf})} (1 - \hat{p}_{m,x}^{(\text{aaf})}) , \quad (\text{A1.10b})$$

and the large sample size estimator is the ratio of averages:

$$F_{ST}(P_x, P_y) = \frac{\frac{1}{M} \sum_{m=1}^M N_m}{\frac{1}{M} \sum_{m=1}^M D_m} . \quad (\text{A1.11})$$

The unbiased estimator of F_{ST} replaces N_m and D_m with bias-corrected versions [2]. The unbiased estimator of the per-site numerator is

$$\hat{N}_m = (\hat{p}_{m,x}^{(\text{aaf})} - \hat{p}_{m,y}^{(\text{aaf})})^2 - \frac{\hat{h}_{m,x}}{c_{m,x}} - \frac{\hat{h}_{m,y}}{c_{m,y}} , \quad (\text{A1.12})$$

and the unbiased estimator of the per-site denominator is

$$\hat{D}_m = \hat{N}_m + \hat{h}_{m,x} + \hat{h}_{m,y} . \quad (\text{A1.13})$$

The unbiased Hudson's F_{ST} estimator is the ratio of averages [2]:

$$\hat{F}_{ST}(P_x, P_y) = \frac{\sum_{m=1}^M \hat{N}_m}{\sum_{m=1}^M \hat{D}_m} . \quad (\text{A1.14})$$

A2 Introgression

Recall that a site pattern is the allelic configuration at a given site for the ordered set $\{P_1, P_2, P_3, O\}$, where the outgroup O is used to polarize ancestral states such that A and B denote the ancestral and derived allele, respectively. For the simplest case of one (haploid) sequence per population, let $P_x(m) \in \{A, B\}$ denote the allele observed in population P_x at site m , and similarly let $O(m)$ denote the outgroup allele. Define indicator variables for each site pattern at site m as

$$\mathbf{1}_{BBAA}(m) = \begin{cases} 1 & \text{if } \{P_1(m) = B, P_2(m) = B, P_3(m) = A, O(m) = A\}, \\ 0 & \text{otherwise,} \end{cases} \quad (\text{A2.1a})$$

$$\mathbf{1}_{BABA}(m) = \begin{cases} 1 & \text{if } \{P_1(m) = B, P_2(m) = A, P_3(m) = B, O(m) = A\}, \\ 0 & \text{otherwise,} \end{cases} \quad (\text{A2.1b})$$

$$\mathbf{1}_{ABBA}(m) = \begin{cases} 1 & \text{if } \{P_1(m) = A, P_2(m) = B, P_3(m) = B, O(m) = A\}, \\ 0 & \text{otherwise,} \end{cases} \quad (\text{A2.1c})$$

$$\mathbf{1}_{BAAA}(m) = \begin{cases} 1 & \text{if } \{P_1(m) = B, P_2(m) = A, P_3(m) = A, O(m) = A\}, \\ 0 & \text{otherwise,} \end{cases} \quad (\text{A2.1d})$$

$$\mathbf{1}_{ABAA}(m) = \begin{cases} 1 & \text{if } \{P_1(m) = A, P_2(m) = B, P_3(m) = A, O(m) = A\}, \\ 0 & \text{otherwise,} \end{cases} \quad (\text{A2.1e})$$

$$\mathbf{1}_{AABA}(m) = \begin{cases} 1 & \text{if } \{P_1(m) = A, P_2(m) = A, P_3(m) = B, O(m) = A\}, \\ 0 & \text{otherwise,} \end{cases} \quad (\text{A2.1f})$$

and given M total sites, the site pattern count is estimated as

$$\hat{n}_{\text{site pattern}} = \sum_{m=1}^M \mathbf{1}_{\text{site pattern}}(m). \quad (\text{A2.2})$$

If more than one sequence per population is available, the site pattern counts can alternatively be computed as

$$\hat{n}_{BBAA} = \sum_{m=1}^M \hat{p}_{m,1}^{(\text{daf})} \hat{p}_{m,2}^{(\text{daf})} (1 - \hat{p}_{m,3}^{(\text{daf})}) (1 - \hat{p}_{m,O}^{(\text{daf})}), \quad (\text{A2.3a})$$

$$\hat{n}_{BABA} = \sum_{m=1}^M \hat{p}_{m,1}^{(\text{daf})} (1 - \hat{p}_{m,2}^{(\text{daf})}) \hat{p}_{m,3}^{(\text{daf})} (1 - \hat{p}_{m,O}^{(\text{daf})}), \quad (\text{A2.3b})$$

$$\hat{n}_{ABBA} = \sum_{m=1}^M (1 - \hat{p}_{m,1}^{(\text{daf})}) \hat{p}_{m,2}^{(\text{daf})} \hat{p}_{m,3}^{(\text{daf})} (1 - \hat{p}_{m,O}^{(\text{daf})}), \quad (\text{A2.3c})$$

$$\hat{n}_{BAAA} = \sum_{m=1}^M \hat{p}_{m,1}^{(\text{daf})} (1 - \hat{p}_{m,2}^{(\text{daf})}) (1 - \hat{p}_{m,3}^{(\text{daf})}) (1 - \hat{p}_{m,O}^{(\text{daf})}), \quad (\text{A2.3d})$$

$$\hat{n}_{ABAA} = \sum_{m=1}^M (1 - \hat{p}_{m,1}^{(\text{daf})}) \hat{p}_{m,2}^{(\text{daf})} (1 - \hat{p}_{m,3}^{(\text{daf})}) (1 - \hat{p}_{m,O}^{(\text{daf})}), \quad (\text{A2.3e})$$

$$\hat{n}_{AABA} = \sum_{m=1}^M (1 - \hat{p}_{m,1}^{(\text{daf})}) (1 - \hat{p}_{m,2}^{(\text{daf})}) \hat{p}_{m,3}^{(\text{daf})} (1 - \hat{p}_{m,O}^{(\text{daf})}), \quad (\text{A2.3f})$$

and if sites are perfectly polarized by the outgroup, then $\hat{p}_{m,O}^{(\text{daf})} = 0$ and $(1 - \hat{p}_{m,O}^{(\text{daf})}) = 1$.

Patterson's D [4] was proposed to detect introgression on a genome-wide scale by comparing patterns of derived allele sharing between P_2 and P_3 to that of P_1 and P_3 , and is estimated as:

$$\hat{D}(P_1, P_2, P_3, O) = \frac{\hat{n}_{ABBA} - \hat{n}_{BABA}}{\hat{n}_{ABBA} + \hat{n}_{BABA}}. \quad (\text{A2.4})$$

D_{anc} [5] was proposed as the ancestral allele counterpart to Patterson's D to detect introgression on both genome-wide and local scales by comparing patterns of ancestral allele sharing between P_2 and P_3 to those of P_1 and P_3 , and is estimated as:

$$\hat{D}_{anc}(P_1, P_2, P_3, O) = \frac{\hat{n}_{BAAA} - \hat{n}_{ABAA}}{\hat{n}_{BAAA} + \hat{n}_{ABAA}}. \quad (\text{A2.5})$$

D^+ [5] was proposed to detect introgression on both genome-wide and local scales by jointly comparing patterns of derived and ancestral allele sharing between P_2 and P_3 to those of P_1 and P_3 , and is estimated as:

$$\hat{D}^+(P_1, P_2, P_3, O) = \frac{(\hat{n}_{ABBA} - \hat{n}_{BABA}) + (\hat{n}_{BAAA} - \hat{n}_{ABAA})}{\hat{n}_{ABBA} + \hat{n}_{BABA} + \hat{n}_{BAAA} + \hat{n}_{ABAA}}. \quad (\text{A2.6})$$

f_{hom} [6] was proposed to quantify the amount of introgression by comparing the observed difference in derived allele sharing between P_3 and P_2 in the numerator to the corresponding value assuming complete homogenization of allele sharing, from P_3 to P_2 in the denominator. In practice, when we satisfy this assumption by replacing P_2 with P_3 in the denominator, the internal branches corresponding to the $BABA$ site pattern are no longer observable, and mutations on the external branch leading to P_3 now contribute to the $ABBA$ site pattern. Accordingly, f_{hom} is estimated as:

$$\hat{f}_{hom}(P_1, P_2, P_3, O) = \frac{\hat{S}(P_1, P_2, P_3, O)}{\hat{S}(P_1, P_3, P_3, O)}, \quad (\text{A2.7})$$

where $\hat{S}(P_1, P_x, P_3, O)$ is estimated as:

$$\hat{S}(P_1, P_x, P_3, O) = \hat{n}_{ABBA} - \hat{n}_{BABA}, \quad (\text{A2.8})$$

for the relevant cases of $x = 2$ and $x = 3$.

D_p [7] was proposed to quantify the amount of introgression from P_3 into P_2 by using all derived allele sharing site patterns, and is estimated as:

$$\hat{D}_p(P_1, P_2, P_3, O) = \frac{|\hat{n}_{ABBA} - \hat{n}_{BABA}|}{\hat{n}_{ABBA} + \hat{n}_{BABA} + \hat{n}_{BBAA}}, \quad (\text{A2.9})$$

note that when performing the weighted block jackknife procedure, the numerator of Eq. (A2.9) is estimated as $\hat{n}_{ABBA} - \hat{n}_{BABA}$.

d_f [8] was proposed to detect introgression on genome-wide and local scales, and to quantify the amount of introgression from P_3 into P_2 , and is estimated as:

$$\hat{d}_f(P_1, P_2, P_3, O) = \frac{\hat{n}_{ABBA} - \hat{n}_{BABA}}{\hat{n}_{ABBA} + \hat{n}_{BABA} + 2\hat{n}_{BBAA}}. \quad (\text{A2.10})$$

D_3 [9] was proposed to leverage pairwise genetic distances to detect introgression on a genome-wide scale. Notably, because D_3 uses pairwise genetic distances instead of site pattern counts, it is the only statistic here that does not rely on an outgroup population and does not require genotypes to be polarized. Accordingly, D_3 is estimated as:

$$\hat{D}_3(P_1, P_2, P_3) = \frac{\hat{d}_{P_1-P_3} - \hat{d}_{P_2-P_3}}{\hat{d}_{P_1-P_3} + \hat{d}_{P_2-P_3}}, \quad (\text{A2.11})$$

note that Eq. (A2.11) can be computed using large sample size form and the unbiased finite sample size estimator of the numerator and denominator.

We propose D^* as an ILS agnostic statistic to detect introgression on a genome-wide scale, and throughout our study, we estimate D^* as

$$\hat{D}^*(P_1, P_2, P_3, O) = \frac{\hat{n}_{ABBA} - \hat{n}_{BABA}}{\hat{n}_{BBAA} + \hat{n}_{ABBA} - 2\hat{n}_{BABA}}. \quad (\text{A2.12})$$

A3 f -Statistics

In this section, P_A, P_B, P_C, P_D, P_E denote generic population labels (not to be confused with the ancestral/derived allele notation A, B from Section A2). Allele frequencies in population P_A are denoted $\hat{p}_{m,A}^{(\text{aaf})}$, and similarly for the other populations. The f -statistics provide unbiased estimates of theoretical F -values averaged across M biallelic segregating sites [10, 11]. For each statistic, we present both the large sample size form and the unbiased finite sample size estimator.

The f_2 statistic quantifies the amount of genetic drift separating populations P_A and P_B , where larger values indicate greater differentiation [10]. The large sample size estimator is

$$f_2(P_A, P_B) = \frac{1}{M} \sum_{m=1}^M (\hat{p}_{m,A}^{(\text{aaf})} - \hat{p}_{m,B}^{(\text{aaf})})^2. \quad (\text{A3.1})$$

The unbiased estimator, which corrects for noise due to finite sample sizes [10, 11], is

$$\hat{f}_2(P_A, P_B) = \frac{1}{M} \sum_{m=1}^M (\hat{p}_{m,A}^{(\text{aaf})} - \hat{p}_{m,B}^{(\text{aaf})})^2 - \frac{\hat{h}_{m,A}}{c_{m,A}} - \frac{\hat{h}_{m,B}}{c_{m,B}}. \quad (\text{A3.2})$$

Note that the correction terms $\hat{h}_{m,A}/c_{m,A}$ and $\hat{h}_{m,B}/c_{m,B}$ are equivalent to $\hat{p}_{m,A}^{(\text{aaf})}(1 - \hat{p}_{m,A}^{(\text{aaf})})/(c_{m,A} - 1)$ when using the unbiased half-heterozygosity from Eq. (A0.5):

$$\frac{\hat{h}_{m,A}}{c_{m,A}} = \frac{n_{m,A}^{(\text{aac})}(c_{m,A} - n_{m,A}^{(\text{aac})})}{c_{m,A}^2(c_{m,A} - 1)} = \frac{\hat{p}_{m,A}^{(\text{aaf})}(1 - \hat{p}_{m,A}^{(\text{aaf})})}{c_{m,A} - 1}.$$

The f_3 statistic quantifies the genetic drift on an unrooted three-population tree separating the target population P_C from the two reference populations P_A and P_B . A significantly negative value suggests that P_C is a mixture of P_A and P_B [10, 11]. The large sample size estimator is

$$f_3(P_C; P_A, P_B) = \frac{1}{M} \sum_{m=1}^M (\hat{p}_{m,C}^{(\text{aaf})} - \hat{p}_{m,A}^{(\text{aaf})})(\hat{p}_{m,C}^{(\text{aaf})} - \hat{p}_{m,B}^{(\text{aaf})}), \quad (\text{A3.3})$$

which equals $\frac{1}{2}[f_2(P_C, P_A) + f_2(P_C, P_B) - f_2(P_A, P_B)]$. The unbiased estimator corrects for the bias introduced because P_C appears in both factors [10, 11]:

$$\hat{f}_3(P_C; P_A, P_B) = \frac{1}{M} \sum_{m=1}^M (\hat{p}_{m,C}^{(\text{aaf})} - \hat{p}_{m,A}^{(\text{aaf})})(\hat{p}_{m,C}^{(\text{aaf})} - \hat{p}_{m,B}^{(\text{aaf})}) - \frac{\hat{h}_{m,C}}{c_{m,C}}. \quad (\text{A3.4})$$

The normalized f_3 statistic divides by the average nucleotide diversity of the target population:

$$\hat{f}_3^*(P_C; P_A, P_B) = \frac{\hat{f}_3(P_C; P_A, P_B)}{2\hat{h}_{P_C}}, \quad (\text{A3.5})$$

where $\hat{h}_{P_C} = \frac{1}{M} \sum_{m=1}^M \hat{h}_{m,C}$ is the average unbiased half-heterozygosity of P_C (Eq. (A0.6b)), and $2\hat{h}_{P_C} = \hat{\pi}_{P_C}$ (Eq. (A1.4)).

The f_4 statistic is a test of “treeness” assuming an unrooted four-population tree $((P_A, P_B), (P_C, P_D))$ where the null hypothesis claims that (P_A, P_B) and (P_C, P_D) form two clades, indicative of a lack of shared genetic drift between populations belonging to different clades. Significant deviations from zero give support for alternative tree topologies, either $((P_A, P_C), (P_B, P_D))$ when positive or $((P_A, P_D), (P_B, P_C))$ when negative, which implies shared genetic drift between populations from different proposed clades and is suggestive of gene flow [10]. Since each population appears in only one factor of the product, no bias correction is needed [10, 11]:

$$\hat{f}_4(P_A, P_B; P_C, P_D) = \frac{1}{M} \sum_{m=1}^M (\hat{p}_{m,A}^{(\text{aaf})} - \hat{p}_{m,B}^{(\text{aaf})})(\hat{p}_{m,C}^{(\text{aaf})} - \hat{p}_{m,D}^{(\text{aaf})}). \quad (\text{A3.6})$$

This is equivalently expressed as a linear combination of f_2 statistics:

$$f_4(P_A, P_B; P_C, P_D) = \frac{1}{2} \left[f_2(P_A, P_D) + f_2(P_B, P_C) - f_2(P_A, P_C) - f_2(P_B, P_D) \right]. \quad (\text{A3.7})$$

Note that correction terms applied to the f_2 statistics in Eqs. (A3.6) and (A3.7) all cancel and thus do not apply to the f_4 statistic.

To test if population P_E has genetic contributions from two source populations, P_B and P_C , and if we know that neither population P_A (closely related to P_B) nor population P_D (an outgroup) contributed to P_E , it is possible to estimate α , the proportion of P_E 's ancestry derived from P_B . The f_4 -ratio statistic estimates this α parameter by taking the quotient of two f_4 statistics on permuted tips [10, 11]:

$$\hat{\alpha} = \frac{\hat{f}_4(P_A, P_D; P_E, P_C)}{\hat{f}_4(P_A, P_D; P_B, P_C)} = \frac{\frac{1}{M} \sum_{m=1}^M (\hat{p}_{m,A}^{(\text{aaf})} - \hat{p}_{m,D}^{(\text{aaf})}) (\hat{p}_{m,E}^{(\text{aaf})} - \hat{p}_{m,C}^{(\text{aaf})})}{\frac{1}{M} \sum_{m=1}^M (\hat{p}_{m,A}^{(\text{aaf})} - \hat{p}_{m,D}^{(\text{aaf})}) (\hat{p}_{m,B}^{(\text{aaf})} - \hat{p}_{m,C}^{(\text{aaf})})}. \quad (\text{A3.8})$$

References

1. Korunes, K. L. & Samuk, K. pixy: Unbiased estimation of nucleotide diversity and divergence in the presence of missing data. *Molecular ecology resources* **21**, 1359–1368 (2021).
2. Bhatia, G., Patterson, N., Sankararaman, S. & Price, A. L. Estimating and interpreting FST: the impact of rare variants. *Genome research* **23**, 1514–1521 (2013).
3. Hudson, R. R., Slatkin, M. & Maddison, W. P. Estimation of levels of gene flow from DNA sequence data. *Genetics* **132**, 583–589 (1992).
4. Green, R. E. *et al.* A draft sequence of the Neandertal genome. *science* **328**, 710–722 (2010).
5. Lopez Fang, L., Peede, D., Ortega-Del Vecchyo, D., McTavish, E. J. & Huerta-Sánchez, E. Leveraging shared ancestral variation to detect local introgression. *PLoS genetics* **20**, e1010155 (2024).
6. Durand, E. Y., Patterson, N., Reich, D. & Slatkin, M. Testing for ancient admixture between closely related populations. *Molecular biology and evolution* **28**, 2239–2252 (2011).
7. Martin, S. H., Davey, J. W. & Jiggins, C. D. Evaluating the use of ABBA–BABA statistics to locate introgressed loci. *Molecular biology and evolution* **32**, 244–257 (2015).

8. Hamlin, J. A., Hibbins, M. S. & Moyle, L. C. Assessing biological factors affecting postspeciation introgression. *Evolution letters* **4**, 137–154 (2020).
9. Hahn, M. W. & Hibbins, M. S. A three-sample test for introgression. *Molecular biology and evolution* **36**, 2878–2882 (2019).
10. Patterson, N. *et al.* Ancient admixture in human history. *Genetics* **192**, 1065–1093 (2012).
11. Reich, D., Thangaraj, K., Patterson, N., Price, A. L. & Singh, L. Reconstructing Indian population history. *Nature* **461**, 489–494 (2009).